Adventure Works 2025 Sales Analytics

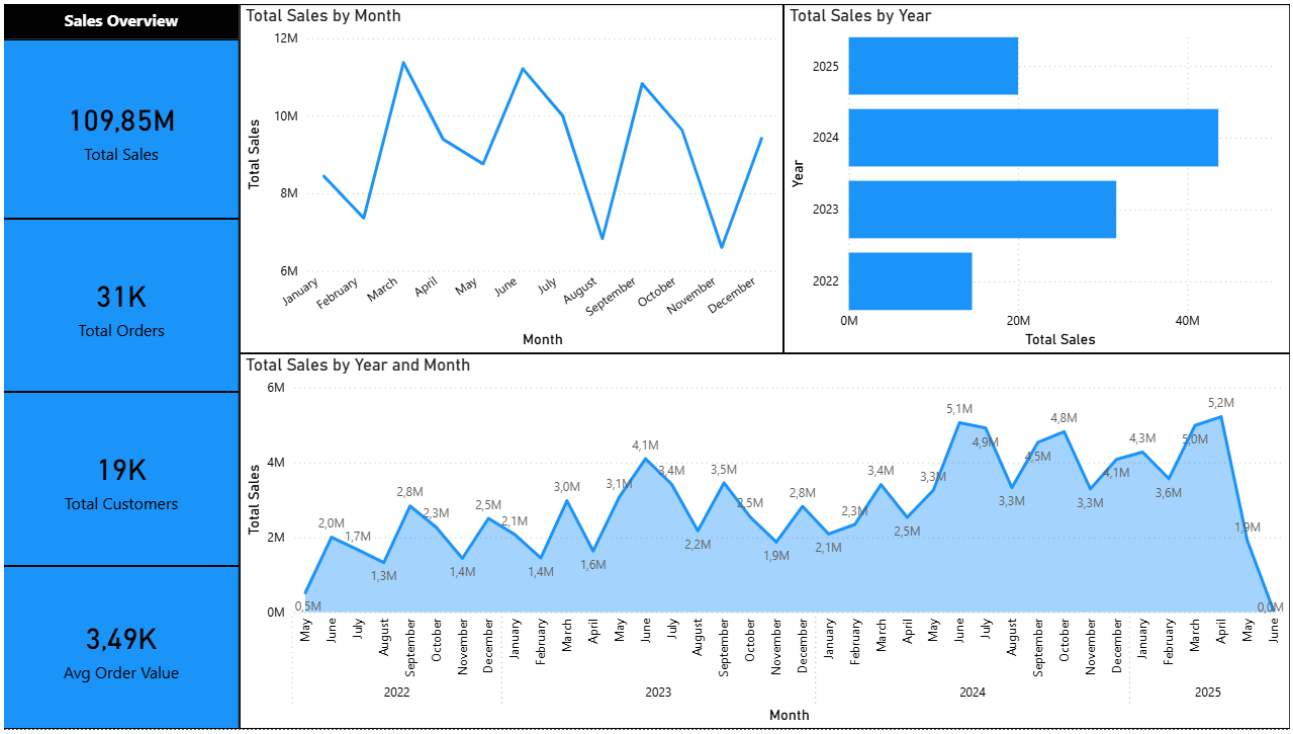
Welcome to the AdventureWorks 2025 Sales Analytics.
This Power BI project provides a deep dive into:

1. Sales overview,
2. Product analysis,
3. Customer analysis,
4. Geographic analysis.

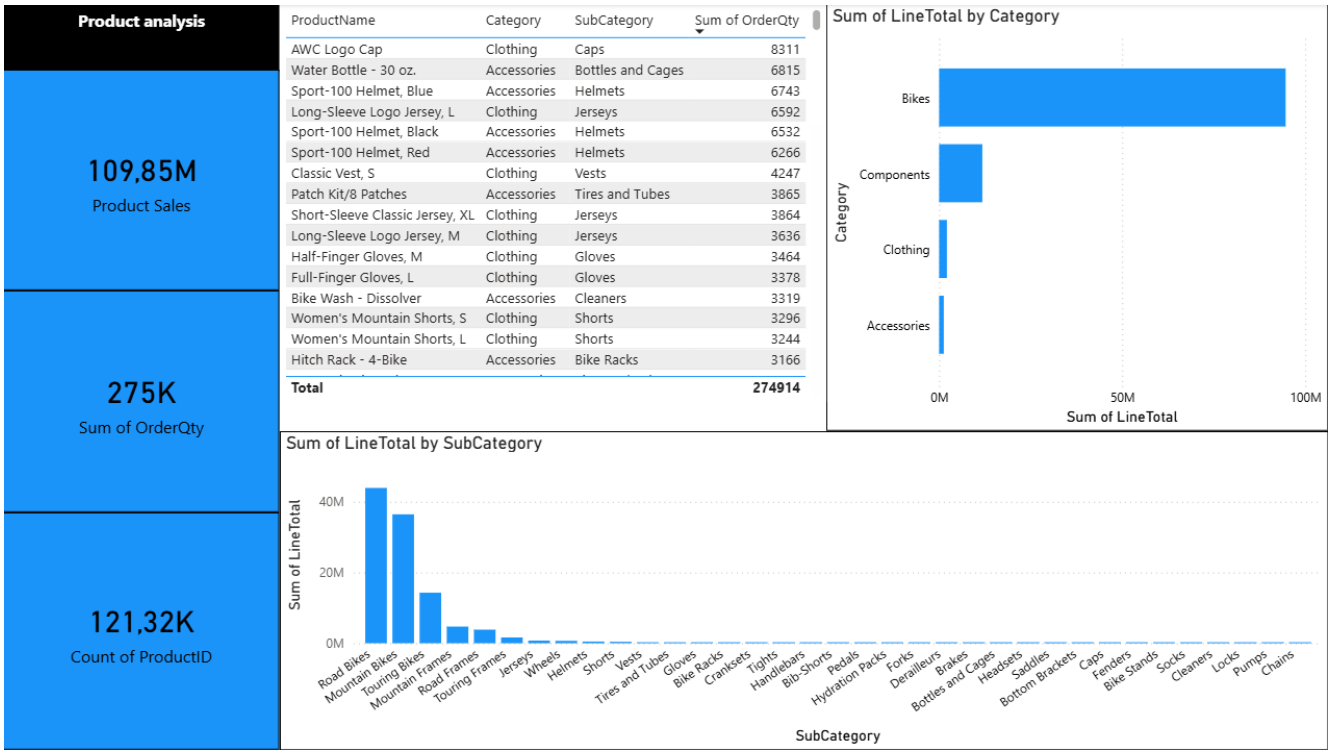
Project Overview

This interactive dashboards analyzes a dataset of product sales between the years 2022 and 2025. It is designed to support business users, analysts, and managers with a comprehensive view of key performance indicators.

1. Sales Overview

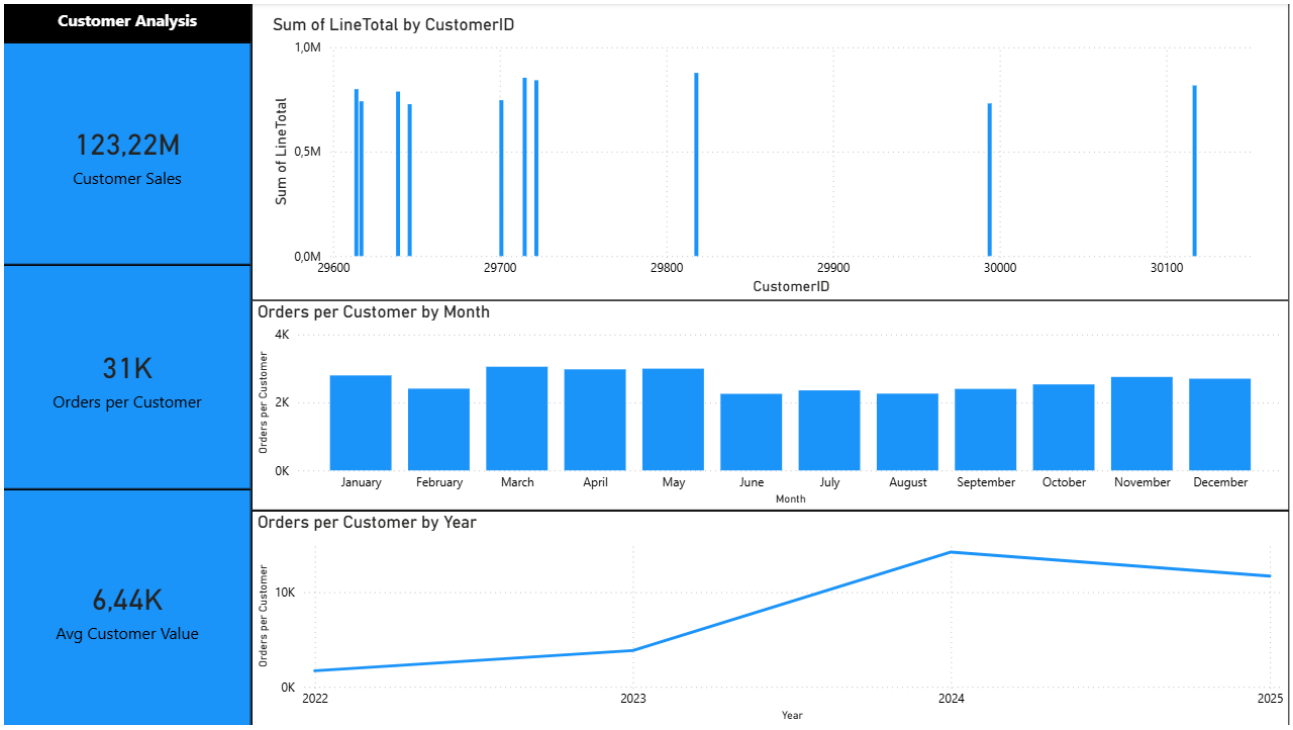


2. Product analysis

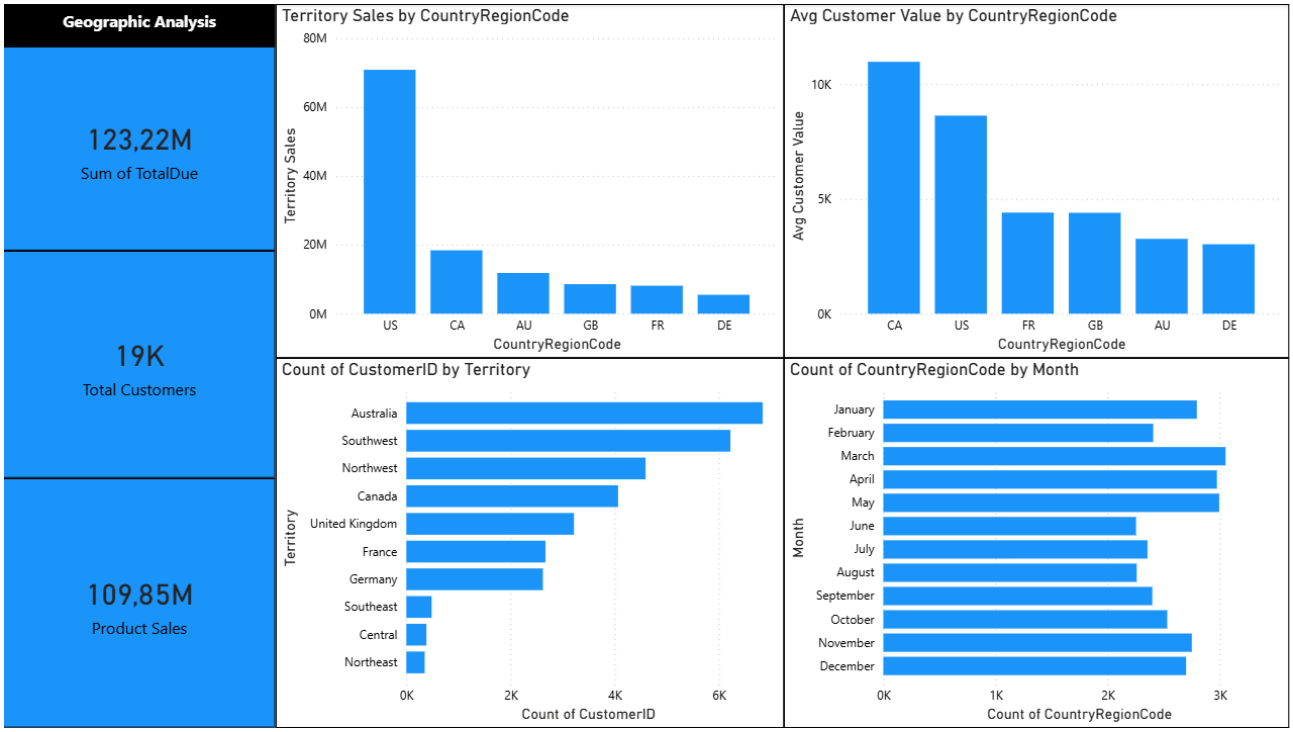


ProductName	Category	SubCategory	Sum of OrderQty
AWC Logo Cap	Clothing	Caps	8311
Water Bottle - 30 oz.	Accessories	Bottles and Cages	6815
Sport-100 Helmet, Blue	Accessories	Helmets	6743
Long-Sleeve Logo Jersey, L	Clothing	Jerseys	6592
Sport-100 Helmet, Black	Accessories	Helmets	6532
Sport-100 Helmet, Red	Accessories	Helmets	6266
Classic Vest, S	Clothing	Vests	4247
Patch Kit/8 Patches	Accessories	Tires and Tubes	3865
Short-Sleeve Classic Jersey, XL	Clothing	Jerseys	3864
Long-Sleeve Logo Jersey, M	Clothing	Jerseys	3636
Half-Finger Gloves, M	Clothing	Gloves	3464
Full-Finger Gloves, L	Clothing	Gloves	3378
Bike Wash - Dissolver	Accessories	Cleaners	3319
Women's Mountain Shorts, S	Clothing	Shorts	3296
Women's Mountain Shorts, L	Clothing	Shorts	3244
Hitch Rack - 4-Bike	Accessories	Bike Racks	3166
Total			**274914**

3. Customer analysis



4. Geographic analysis



Key Questions Addressed:

- What are the total sales and profits?
- Which products sell the most?
- Which months and years generate higher sales?
- What are most popular categories and subcategories?
- Who are top 10 customer?
- What are the sales in regions and territory?

Key Metrics:

- **Total Sales: $109.85M**
- **Total Orders: 31K**
- **Total Customer: 19K**
- **AVG Order Value: 5.75K**

- **Sum of OrderQty: 275K**
- **Count of ProductID: 121.32K**
- **Customer Sales: 123.22M**
- **AVG Customer Value: 6.44K**
- **Sum of TotalDue: 123.22M**

Highlights & Insights:

Top Selling Product:

Bikes

Top selling subcategory:

Road Bikes

Product Quantity Ordered:

Bikes – 94 651 172. 70

Components – 11 802 593. 29

Clothing – 2 120 542. 52

Accessories – 1 272 072.88

Year-wise Sales Trend

- Year 2024 had the highest spike in sales (43.47M),
- Lowest sales observed in 2022 (14.56M).

Month-wise Sales Trend

- Month April 2025 had the highest spike in sales (5.2M),
- Lowest sales observed in August 2022 (1.3M).

Tools and Technologies Used:

- **Power BI Desktop** – Dashboard development
- **DAX** – Calculated fields for KPIs
- **SQL Server** – Source data formatting and transformation

1. Sales Overview

SQL QUERY:

```
SELECT
   soh.OrderDate,
   soh.SalesOrderID,
   soh.CustomerID,
   soh.TotalDue,
   sod.OrderQty,
   sod.LineTotal
FROM Sales.SalesOrderHeader soh
JOIN Sales.SalesOrderDetail sod
   ON soh.SalesOrderID = sod.SalesOrderID
```

Measures (DAX):

```
Total Sales = SUM(Query1[LineTotal])
Total Orders = DISTINCTCOUNT(Query1[SalesOrderID])
Total Customers = DISTINCTCOUNT(Query1[CustomerID])
Avg Order Value = DIVIDE([Total Sales], [Total Orders])
```

2. Product analysis

SQL QUERY:

```
SELECT
   p.ProductID,
   p.Name AS ProductName,
   pc.Name AS Category,
   psc.Name AS SubCategory,
   sod.OrderQty,
   sod.LineTotal
FROM Production.Product p
JOIN Production.ProductSubcategory psc
   ON p.ProductSubcategoryID = psc.ProductSubcategoryID
JOIN Production.ProductCategory pc
   ON psc.ProductCategoryID = pc.ProductCategoryID
JOIN Sales.SalesOrderDetail sod
   ON p.ProductID = sod.ProductID
```

3. Customer analysis
4. Geographic analysis

Measures (DAX):

```
Product Sales = SUM(Query1[LineTotal])
```

Quantity Sold = SUM(Query1[OrderQty])

3. Customer analysis

SQL QUERY:

```
SELECT
    c.CustomerID,
    p.FirstName,
    p.LastName,
    soh.SalesOrderID,
    soh.OrderDate,
    soh.TotalDue
FROM Sales.Customer c
JOIN Person.Person p
    ON c.PersonID = p.BusinessEntityID
JOIN Sales.SalesOrderHeader soh
    ON c.CustomerID = soh.CustomerID
```

Measures (DAX):

Customer Sales = SUM(Query3[TotalDue])
Orders per Customer =DISTINCTCOUNT(Query3[SalesOrderID])
Avg Customer Value = DIVIDE([Customer Sales], DISTINCTCOUNT(Query3[CustomerID]))

4. Geographic analysis

SQL QUERY:

```
SELECT
    st.Name AS Territory,
    st.CountryRegionCode,
    soh.SalesOrderID,
    soh.TotalDue,
    soh.OrderDate
FROM Sales.SalesOrderHeader soh
JOIN Sales.SalesTerritory st
    ON soh.TerritoryID = st.TerritoryID
```

Measures (DAX):

Territory Sales = SUM(Query4[TotalDue])